COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
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NAME:	Momo, Borge Rudolf A.	**INSTRUCTOR:**	Engr. Quejado

PRELIM SKILLS EXAM

1. PROCEDURE OUTPUT

Step 1 Set Up

Hostname	Operating System	IP Address
momo@server1	Ubuntu	192.168.56.115
momo@server2	Ubuntu	192.168.56.114
borj@centos	CentOS	192.168.56.116

Part A Installing and Verifying Ansible

1. Update the package repository.
2. Install pip if it is not yet installed.
3. Install Ansible.
4. Verify that Ansible is installed correctly.
5. Verify SSH connectivity to each managed node

Ansible Version

```
momo@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/momo/.ansible/plugins/modules', '/usr/
share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/momo/.ansible/collections:/usr/share/ansib
le/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
momo@workstation:~$
```

SSH Connectivity

Control Node to Ubuntu1 (momo@server1)

```
momo@workstation:~$ ssh momo@192.168.56.115
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

97 updates can be applied immediately.
79 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 16 01:19:37 2026 from 192.168.56.113
momo@server1:~$
```

Control Node to Ubuntu2 (momo@server2)

```
momo@workstation:~$ ssh momo@192.168.56.114
The authenticity of host '192.168.56.114 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:FwwF11OhlUOIdn8ES3Xn8kXZmqtX2Dy7/vnhfNL23Xk.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.114' (ED25519) to the list of known hosts
.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Thu Jul 16 01:20:00 2026 from 192.168.56.113
momo@server2:~$
```

Control Node to CentOS (borj@centos)

```
momo@workstation:~$ ssh borj@192.168.56.116
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Aug 13 08:11:39 2026 from 192.168.56.113
[borj@centos ~]$
logout
Connection to 192.168.56.116 closed.
```

Demonstrate the following:

Inventory.yaml

```
  GNU nano 7.2                                inventory.yaml
all:
  children:

    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.115

        ubuntu2:
          ansible_host: 192.168.56.114

  centos:
    hosts:
       centos1:
          ansible_host: 192.168.56.116
```

Successful inventory verification

```
momo@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "centos1": {
                "ansible_host": "192.168.56.116"
            },
            "ubuntu1": {
                "ansible_host": "192.168.56.115"
            },
            "ubuntu2": {
                "ansible_host": "192.168.56.114"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu",
            "centos",
            "linuxservers"
        ]
    },
    "centos": {
        "hosts": [
            "centos1"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu1",
            "ubuntu2"
```

Successful ping results

```
momo@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
momo@workstation:~/ansible-lab$
```

Part C: Creating a Metagroup

Demonstrate successful execution of the metagroup.

```
momo@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yam
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
momo@workstation:~/ansible-lab$
```

Part D: Creating Variables

Demonstrate successful use of variables.

```
momo@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] ***********************************************************

TASK [Gathering Facts] *************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Show Operating System] *******************************************************
ok: [ubuntu1] => {
    "msg": "Host ubuntu1 is running Ubuntu"
}
ok: [ubuntu2] => {
    "msg": "Host ubuntu2 is running Ubuntu"
}
ok: [centos1] => {
    "msg": "Host centos1 is running CentOS"
}

PLAY RECAP *************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0
   rescued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0
   rescued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0
   rescued=0    ignored=0

momo@workstation:~/ansible-lab$
```

Part E: Creating Your First Playbook

Successful playbook execution

```
momo@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [First Ansible Playbook] **************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Ping all managed nodes] **************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Display Hello World] *****************************************************
ok: [ubuntu1] => {
    "msg": "Hello World from ubuntu1"
}
ok: [ubuntu2] => {
    "msg": "Hello World from ubuntu2"
}
ok: [centos1] => {
    "msg": "Hello World from centos1"
}

TASK [Display Operating System] ************************************************
ok: [ubuntu1] => {
    "msg": "Operating System: Ubuntu"
}
ok: [ubuntu2] => {
    "msg": "Operating System: Ubuntu"
```

Successful task completion on all managed nodes

```
PLAY RECAP *********************************************************************
centos1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named servers that contains only:

- ubuntu1
- centos1

```
  GNU nano 7.2                                servers.yaml
all:
  children:
    ubuntu:
      vars:
        ansible_user: momo
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.115
          os_type: Ubuntu
    centos:
      vars:
         ansible_user: momo
      hosts:
         centos1:
          ansible_host: 192.168.56.116
          os_type: CentOS
    servers:
      children:
        ubuntu:
        centos:
```

Create a playbook named challenge.yaml that displays the message:

```
  GNU nano 7.2                          challenge.yaml *
---
- name: display a message from supplementary activity hehe
  hosts: servers

  tasks:
    - name: display a cute message
      debug:
        msg: "Welcome to Ansible Laboratory, {{ inventory_hostname }}"
```

Run the playbook and demonstrate that only hosts in the servers group receive the message.

```
momo@workstation:~/ansible-lab$ ansible-playbook -i servers.yaml challenge.yaml

PLAY [display a message from supplementary activity hehe] ***********************
*******

TASK [Gathering Facts] ***************************************************
*******
ok: [ubuntu1]
ok: [centos1]

TASK [display a cute message] ********************************************
*******
ok: [ubuntu1] => {
    "msg": "Welcome to Ansible Laboratory, ubuntu1"
}
ok: [centos1] => {
    "msg": "Welcome to Ansible Laboratory, centos1"
}

PLAY RECAP ***************************************************************
*******
centos1                    : ok=2    changed=0    unreachable=0    failed=0    sk
ipped=0    rescued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    sk
ipped=0    rescued=0    ignored=0

momo@workstation:~/ansible-lab$
```

Guide Questions

1. What is the role of the Control Node in Ansible?
   - The Control Node is basically the administrator or the workstation to the remote servers. It is the one that handles, modifies and controls the remote servers.
2. Why is an inventory file required?
   - An inventory file is important because it is where the Ansible will know which servers or hosts to manage.
3. What is the purpose of a metagroup?
   - A metagroup is used to group other groups of hosts together. This makes it easier to run the same tasks on multiple groups at once.
4. How do variables simplify Ansible administration?
   - It allows us to store values that can be reused in different tasks and playbooks.
5. What is the difference between an ad hoc command and a playbook?
   - The ad hoc command is used to perform a single task on remote servers. A playbook is used to perform multiple tasks

CONCLUSIONS

In conclusion, I learned a lot from this Prelim Skills Exam. It taught me that I was able to work on this activity independently. Although it was hard at first and I was stuck at a part for almost 30 minutes, I still managed to debug the errors and perform all the procedures and the supplementary activity.

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